

EECS 300 Semester-Long Project

Project Overview

Each team will design, build, test, and validate a system to collect and process data for a specific purpose. The objective of this semester is to design a system that either 1) counts the number of people inside a room in real-time (**the default project**) or 2) performs a “useful” function defined by the team and approved by the instructor (**a team-defined project**). This project will be “Internet of Things (IoT)” focused, in that it will require these systems to communicate with a server via a wireless communication channel. Each team can determine whether to perform the default project or a team-defined project. If a team selects a team-defined project, they must clearly specify the system’s functionality and evaluation criteria, and obtain approval from the instructor.

Option 1) Default Project: Room Occupancy Monitoring

The sensing objective is to accurately count and estimate the number of people in a given space. The system must support a room with a capacity of up to 100 people. The following are assumed / required:

- Wireless operation with batteries and WiFi connection
- ESP32 should be the main platform. At least two ESP32’s are used for the system: one or more units for sensing and one unit for the server to report the people count.
- Support regular human traffic as well as corner cases defined by the instructor
- The testing door (installed in the lab) allows up to 2 people to pass through side by side in the same and opposite directions
- The door should be able to close with the system installed.

Option 2) Team-defined Project

The system functionality and objective can be defined by each team for a team-defined project. Each team must determine the evaluation criteria and get approval from the instructor before / during the design review. The team has the freedom to define the system. However, it must perform a **useful function that has a positive societal impact**, such as a wireless head-mount mouse for people with hand disabilities (similar

to [this](#) example, augmented by another ESP32 unit to add the touch-click feature). Avoid proposing a system whose primary function is entertainment or a toy, unless it is specifically designed to support people with disabilities.

The following are assumed / required:

- Wireless operation with a battery and WiFi connection
- Must use one or more sensors
- ESP32 should be the main platform. At least two ESP32's are used for the system. In addition, a PC can be used as a server WiFi connected to ESP32 units. For example, a mouse cursor wirelessly controlled by the ESP32's can be shown on a server PC for demonstration, as in this [example](#).
- Evaluation criteria must be quantifiable and clearly defined by the team, and approved by the instructor

At the end of the semester, there will be a project demonstration, where each team will present a system and show it working based on either the standardized testing procedure (default project) or the user-defined (and instructor-approved) evaluation criteria.

Constraints

Each project will be given a budget, for both testing and final assembly. The bill of materials (BOM) for the final prototype must be less than \$100 per unit. **The BOM is how much you would need to budget when you mass-produce your devices from a contract manufacturer. It does not need to include labor costs, manufacturing overheads such as electricity, or components that you may have used during the R&D but are not part of your final prototype. However, it does need to include the actual cost of the components even if they have been given to you for free during R&D, e.g., the ESP32.** When calculating the BOM, you may assume that at scale, you are ordering thousands of each part for price discounts. (Hence, it is different from the budget that you spend to produce **a single unit** of the prototype system you demo at the end of the semester.)

Final projects can consist of a single, higher-fidelity sensor package or multiple different sensor packages distributed throughout a relevant area that communicate with each other as needed. This choice will be constrained by the project budget. The communication between the sensor and the server must be wireless.

It is strongly recommended that ESP32 be used as the main platform for final projects. This will make it easier for course staff to help students debug. **Using other platforms requires approval from the course instructor.** The staff support for those platforms may NOT be provided.

Each team's final prototype must also include a PCB that the team designed. Getting this done early is key to sorting out any issues and leaving time to order a second one if needed. **All teams must have their PCB design reviewed by the GSI/IA before ordering.**

Power Management

The final prototype must be battery-operated. Teams are also required to analyze the expected battery lifetime and provide data to support the claim.

Budget

Each team will have a \$200 budget for R&D and prototype system development. This is the actual money you may spend during the project to get the prototype working. It can include the purchasing of various components, including your PCBs. Some components will already be in stock in course storage and you do not need to purchase them (so they will **not** be counted toward your budget). **In other words, \$200 counts only the dollars your team spends throughout the project.** For each purchase order, there will be a \$5 surcharge (see the Resources Section below for more details). **Although you can spend up to \$200 for your R&D, the bill of materials (BOM) for the final prototype must be less than \$100 for each unit.**

Overspending the budget is not allowed and will lead to a grade penalty.

Resources

To procure parts for your system, please either submit a purchase order (PO) through EECS or buy a part yourself and get reimbursed at the end of the semester (Receipts should be saved and turned in for reimbursement). All purchases must be documented in your project budget. To submit a PO use this link:

<https://www.eecs.umich.edu/eecs/etc/pos/form.html>.

Once you're in the OPS website, fill out the following information and your order details. **Make sure you click "Submit Request" at the bottom of the webpage.** It usually takes a few hours (during business hours) to a day for the Department to place the order for you. However, you should follow up with the vendor to check the lead time. You should also check the status of your order by using the OPS website to make sure nothing falls through the cracks.

Online Purchasing System (OPS)

This is the new ECE Purchasing System. Here are instructions for placing an order in the new system: [ECE OPS Tutorial](#). You can see an overview of the process on the [Site Info](#) page.

Purchase Request Form New [Change Department](#)

Basic Information

Order Nickname

Order placed by: Ku, Pei-Cheng (peicheng)

☒ Order placed on someone else's behalf

Order placed on behalf of: not set [Notify on submission](#)

Start typing name or unique name...

Course Number **EECS 300**

Purpose

Purchase electronic components as part of the class project.

☐ This is a hosting request

☐ This request has already been ordered

☐ This is a chemical/gas/powder/solvent purchase request

☐ This is a server, workstation, laptop, printer, or tablet purchase request

Shortcodes

You are not required to enter shortcodes in order to submit a request. However, if you do not specify shortcodes, you must provide the username of a Shortcode Owner who will then provide the appropriate shortcode(s).

Shortcode	Shortcode Owner	Percentage	Amount
182093	(not set)	100 %	\$ 0.00

Shortcode is required

[ADD SHORTCODE](#)

Shipping

Recipient: **Your name**

Shipping Address: **EECS Dock**

Room #

☐ Expedited shipping

Supplier Information

You must specify a supplier.

Search for a Supplier

Start typing supplier name

☐ Use unlisted supplier

Check (points to "Order placed on someone else's behalf")

GSI unique name: bwyang or aidaruan (points to "Start typing name or unique name...")

Select EECS Dock (points to "Shipping Address: EECS Dock")

Choose the vendor (points to "Search for a Supplier")

Select "Add Shortcode" (points to "ADD SHORTCODE")

The amount will be automatically updated once you fill out the order details. (points to "Amount" column in Shortcodes table)

Note that PO's can only be for one vendor. We will be applying an artificial \$5 surcharge (meaning that it comes out of your prototyping budget) to each order whether you go through EECS or purchase yourself. This is to encourage you to plan well and minimize the number of your orders, as well as to order from fewer different vendors.

Main Deliverables

Design Review 1 Presentation ([link](#) to more information)

Teams will present a set of **three** initial ideas as final project candidates. All three ideas can be for the default people counting system, or you can include one or more user-defined systems. Select one idea for your final project and briefly justify why you chose that idea. The scope of each idea in this presentation is a basic theoretical analysis of how the idea would accomplish the provided project sensing objective and a prediction of how effective it would be at doing so. Pros and cons of each suggested idea should also be discussed. The divergent/convergent design process should be described getting to this point.

Design Review 2 Presentation ([link](#) to more information)

Teams will present an in-depth overview of their selected design idea. This will include:

- A complete system plan with a functional decomposition description, including the purpose of each subsystem and how they connect/communicate
- A plan for developing this idea in the remainder of the semester. This typically includes a Gantt chart, but focus should be put on the critical path and key milestones with planned tests to demonstrate milestone achievement. Individual responsibilities should also be clearly identified.
- A deeper analysis of the predicted performance of the system than in Design Review 1
- A list of anticipated challenges or risks and how they will be addressed

Milestones 1 and 2

Between the two design reviews and between the design review 2 and final demonstration, there will be two project checkpoints. These checkpoint demonstrations will be scheduled during the regular lab hours. For more information, click the following links.

[Milestone 1](#)

[Milestone 2](#)

Physical Prototype ([link](#) to final demonstration)

Teams will construct a final physical prototype of their system. Part of their project grade will be based on the actual demonstrable performance of their system.

Final Paper ([link](#) to more information)

Teams will submit a paper at the end of the semester, which has the following objectives:

- Analyze team performance
- Allow readers to re-create a team's work (technical documents should be in the appendix or in an associated zip file)
- Discuss the performance of the final Physical Prototype
- Offer suggestions for the continuation of the final project
- Help readers understand the reasoning behind a team's design decisions (these should be meaningful and well thought out)

Weekly Updates

Part of what makes a project happen goes beyond the technical skills of the engineering team and depends on being focused on the goal, organized, and proactive at resolving issues - in short, having actual project management. Each team will be required to select a project manager who will be responsible for keeping the team on track and submitting weekly update assignments. These assignments will be submitted to the GSIs and the Faculty instructor(s). Please submit these updates in the Google Drive folder assigned to your team. **Please contact the instructors if you face significant collaboration issues (such as a team member being mostly absent without making any meaningful contributions) that prevent making steady progress.**

Each weekly update should contain

1. What each team member did over the past week (2-5 sentences per team member), including how many hours worked on the project (not including regular lab assignment time).
2. Identifying if a team member is currently blocked and needs to resolve to move ahead (1-2 sentences).
3. What each team member is planning to do over the next 2-3 weeks, taking into account that individual's scheduling constraints (interviews; travel; etc.).

4. A brief summary of the overall project status, critical path risks, and plans to mitigate any issues (less than 10 sentences).
5. A table summarizing the team finance (purchases including \$5 surcharges, current balance, projected balance at the end of the project).

The weekly updates will be graded only on your ability to answer the questions, not on their content (i.e. you won't be penalized for the amount of progress on your project, but you will be penalized for not having a clear personal plan for the next two weeks, or not reporting obstacles that clearly exist). Note that the Weekly Updates are graded as part of the Project Management section of Class Grades (See Syllabus).

Grading

Details can be found for each assignment's specific requirements on Canvas.

Assignment	Points (out of 100)
Design Review 1	15
Design Review 2	15
Final Demonstration	20
Final Paper	20
Project Checkpoint Milestone 1	15
Project Checkpoint Milestone 2	15